

Problem rating sheet

Rater: _____ Date: _____

Why this exists. I am building software that writes new contest-style problems, from AMC 10 level up to AIME level, plus a few quant-style probability problems. Before I trust it, I need one expert judgement of what a good problem looks like. The 14 problems below mix problems from past contests with machine-generated ones, in an order chosen so that neither kind is grouped, and with no labels. Please don't try to guess the source, and if you recognise a problem, rate it as you would any other.

Time. About 30 to 45 minutes; the sheet is two sides. You need not solve every problem completely: read each one, solve or sketch the ones you can, and skim the rest. No answer key is attached. Imagine contest conditions: no calculator, free response (no answer choices).

How to mark. For each problem, circle one number from 1 to 5 for each question (1 = no, 5 = definitely):

- **Q1 Well-posed.** Is it unambiguous, with a single correct answer of the stated form?
- **Q2 Difficulty.** Is the difficulty right for the contest and position you wrote under "Feels like"? (5 = about right for that slot; 1 = badly mismatched, too easy or too hard.)
- **Q3 Interesting.** Is it a problem worth solving, rather than a mechanical routine?
- **Q4 Would use.** Would you give it to students?

"Feels like" is optional free text: roughly which contest and position the problem feels like (for example AMC 10 #15, AMC 12 #22, AIME #6). "Comment" is optional: an ambiguity, a different answer, a suggested fix, anything. Leave blank whatever you would rather skip.

1. It takes Clea 60 seconds to walk down an escalator when it is not operating, and only 24 seconds to walk down the escalator when it is operating. How many seconds does it take Clea to ride down the operating escalator when she just stands on it? *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

2. A fair 6-sided die, with faces 1 to 6, is rolled repeatedly until the running total exceeds 3. What is the expected number of rolls? *Answer format: a common fraction in lowest terms.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

3. The solutions to the equation $\log_{3x} 4 = \log_{2x} 8$, where x is a positive real number other than $\frac{1}{3}$ or $\frac{1}{2}$, can be written as $\frac{p}{q}$ where p and q are relatively prime positive integers. What is $p + q$? *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

4. Find the remainder when 3^{2^4} is divided by 500. *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

5. Quadrilateral $ABCD$ is inscribed in a circle with $\angle BAC = 70^\circ$, $\angle ADB = 40^\circ$, $AD = 4$, and $BC = 6$. What is AC ? *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

6. How many ways are there to split the integers 1 through 14 into 7 pairs such that in each pair, the greater number is at least 2 times the lesser number? *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

7. Find the remainder when 12^{17^8} is divided by 990. *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

8. Three numbers in the interval $[0, 1]$ are chosen independently and at random. What is the probability that the chosen numbers are the side lengths of a triangle with positive area? *Answer format: a common fraction in lowest terms.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

9. A fair 4-sided die, with faces 1 to 4, is rolled repeatedly until the running total exceeds 5. What is the expected number of rolls? *Answer format: a common fraction in lowest terms.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

10. Let x and y be positive integers such that $7x^5 = 11y^{13}$. The minimum possible value of x has a prime factorization $a^c b^d$. What is $a + b + c + d$? *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

11. Find the remainder when 11^{12^4} is divided by 700. *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

12. Three balls are randomly and independently tossed into bins numbered with the positive integers so that for each ball, the probability that it is tossed into bin i is 2^{-i} for $i = 1, 2, 3, \dots$. More than one ball is allowed in each bin. The probability that the balls end up evenly spaced in distinct bins is $\frac{p}{q}$, where p and q are relatively prime positive integers. (For example, the balls are evenly spaced if they are tossed into bins 3, 17, and 10.) What is $p + q$? *Answer format: an integer from 0 to 999.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

13. A fair 6-sided die, with faces 1 to 6, is rolled repeatedly until the running total exceeds 7. What is the expected number of rolls? *Answer format: a common fraction in lowest terms.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	

14. The number 2017 is prime. Let $S = \sum_{k=0}^{62} \binom{2014}{k}$. What is the remainder when S is divided by 2017? *Answer format: a nonnegative integer.*

Feels like	Q1 Well-posed	Q2 Difficulty	Q3 Interesting	Q4 Would use	Comment
	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	